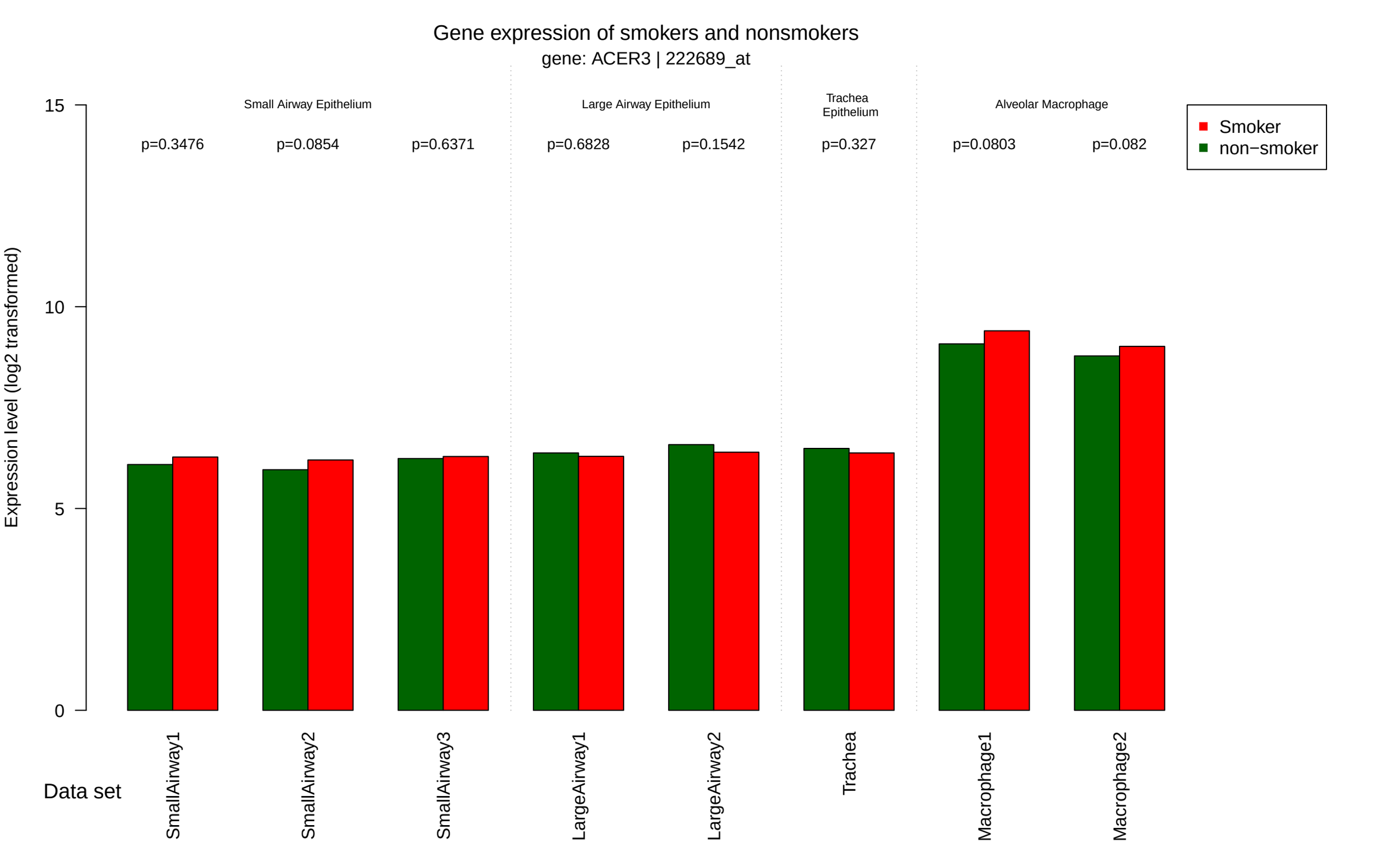
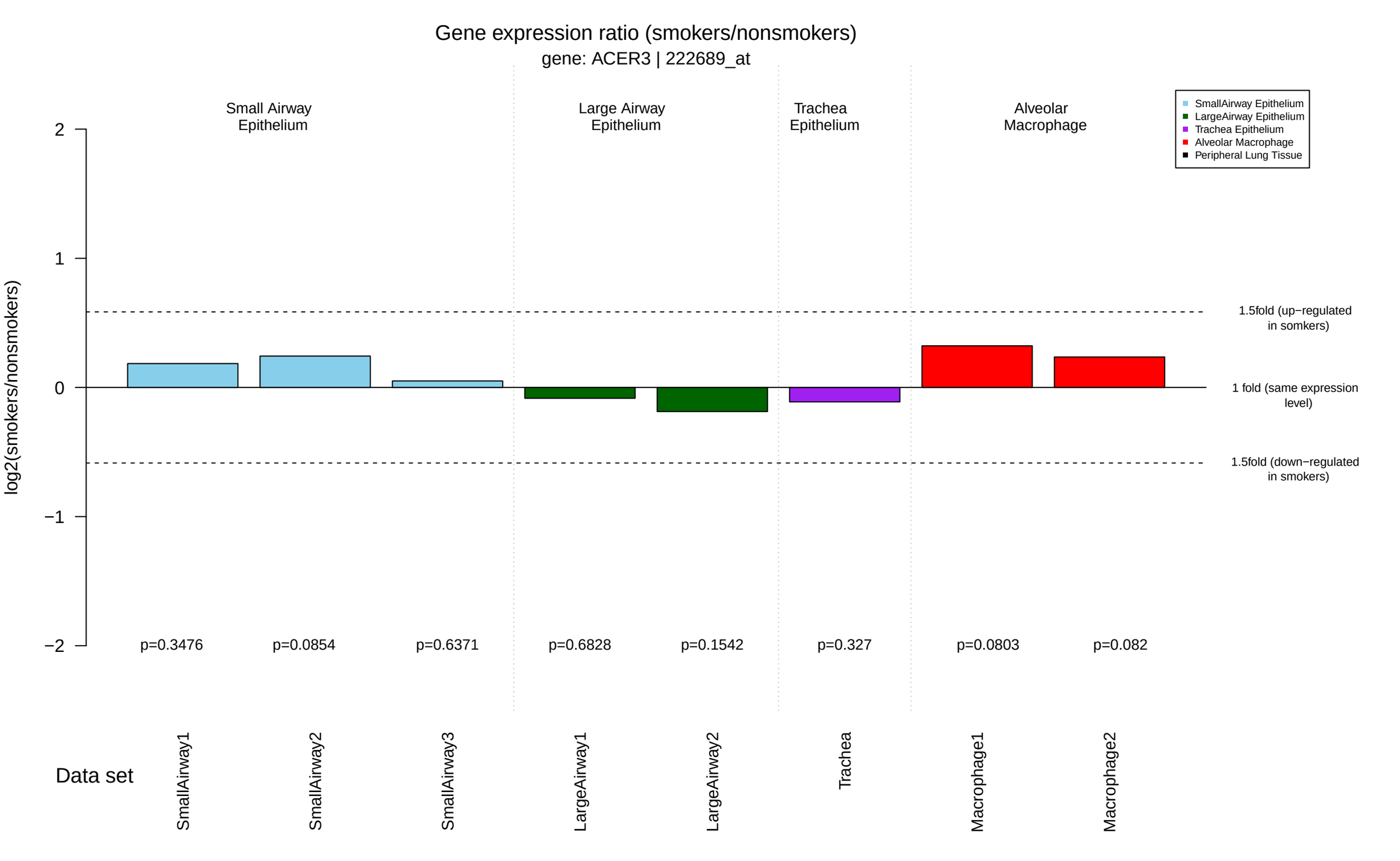
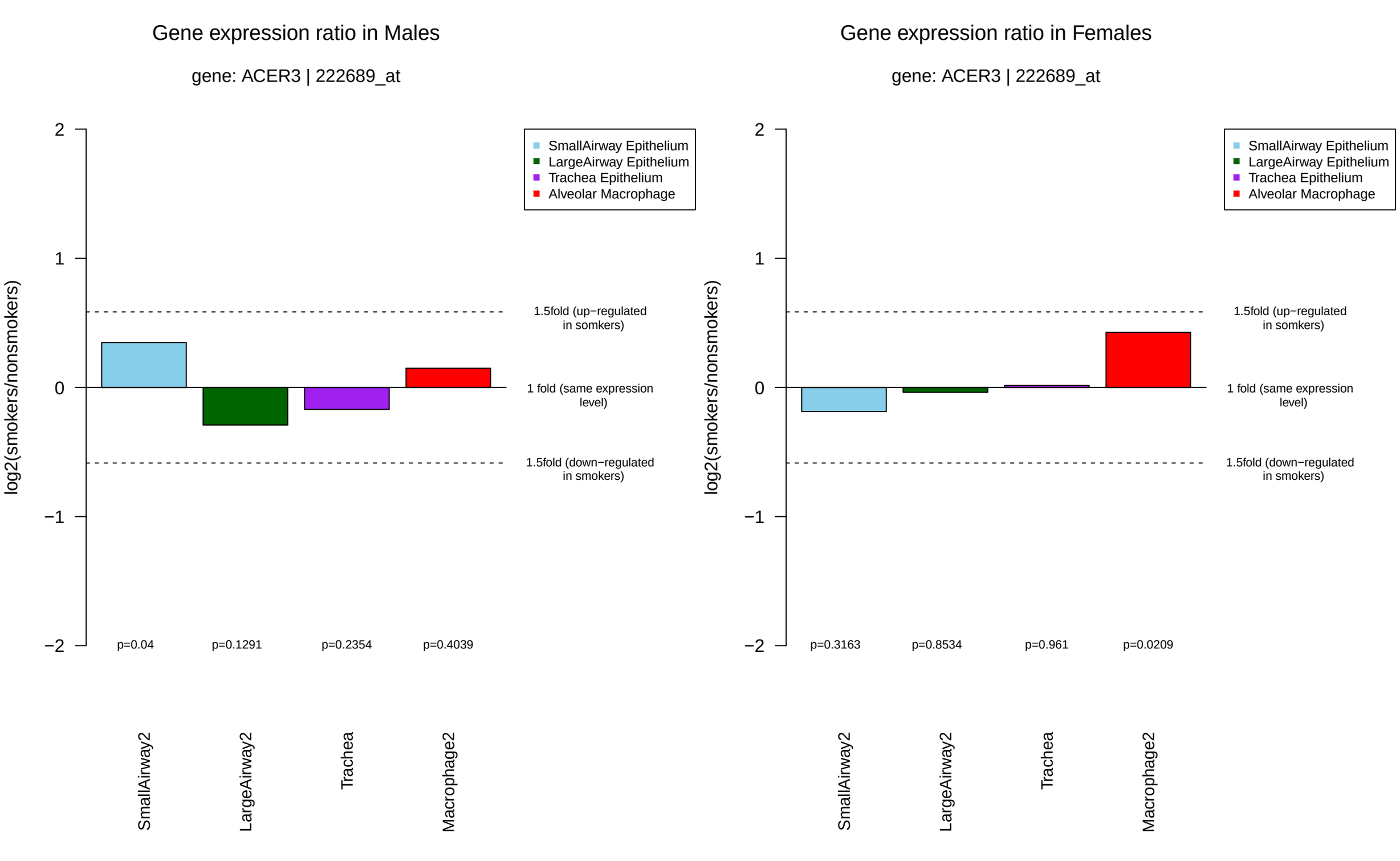
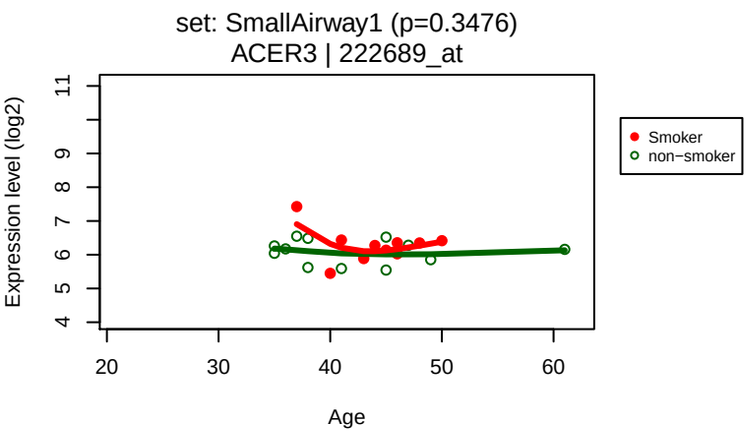
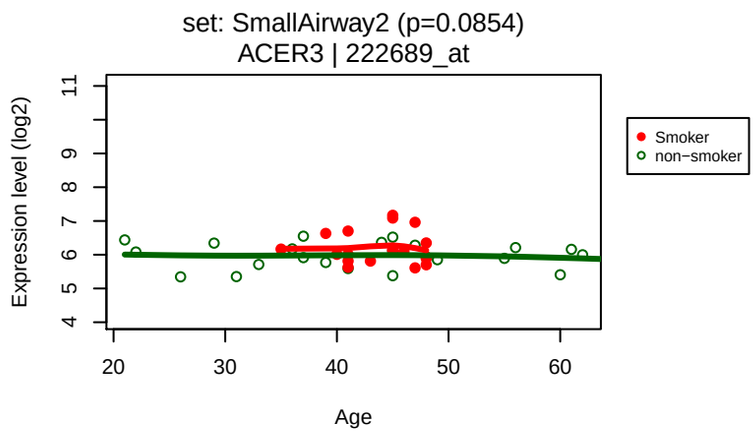
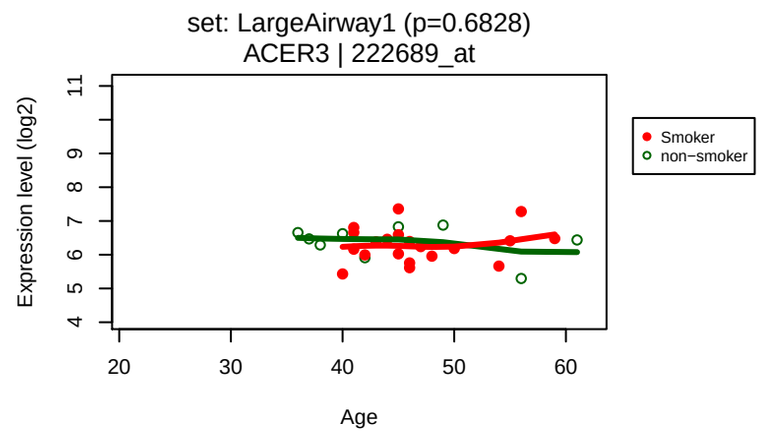

Gene expression of smokers and nonsmokers
gene: ACER3 | 222689_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3476
p=0.0854
p=0.6371
p=0.6828
p=0.1542
p=0.327
p=0.0803
p=0.082
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

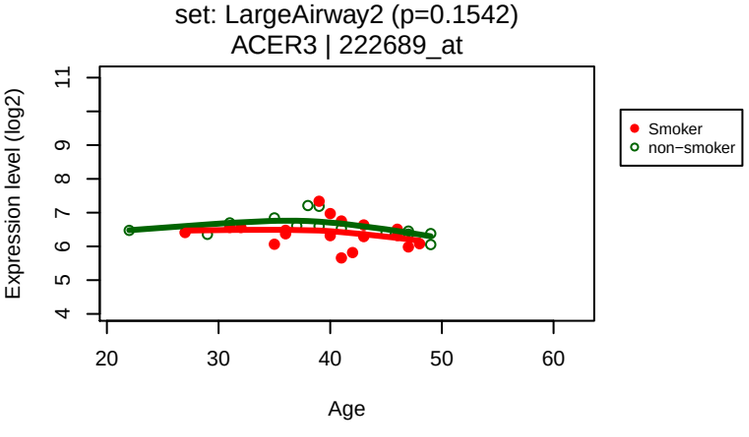

Gene expression ratio (smokers/nonsmokers)
gene: ACER3 | 222689_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3476
p=0.0854
p=0.6371
p=0.6828
p=0.1542
p=0.327
p=0.0803
p=0.082
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

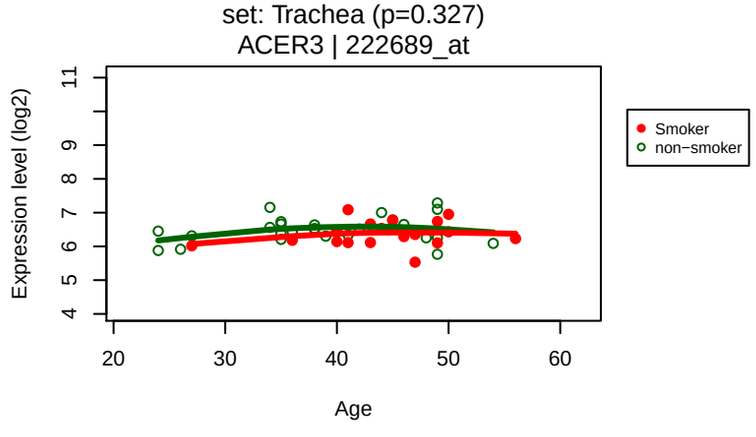

Gene expression ratio in Males
gene: ACER3 | 222689_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.04
p=0.1291
p=0.2354
p=0.4039
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACER3 | 222689_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3163
p=0.8534
p=0.961
p=0.0209
SmallAirway2
LargeAirway2
Trachea
Macrophage2

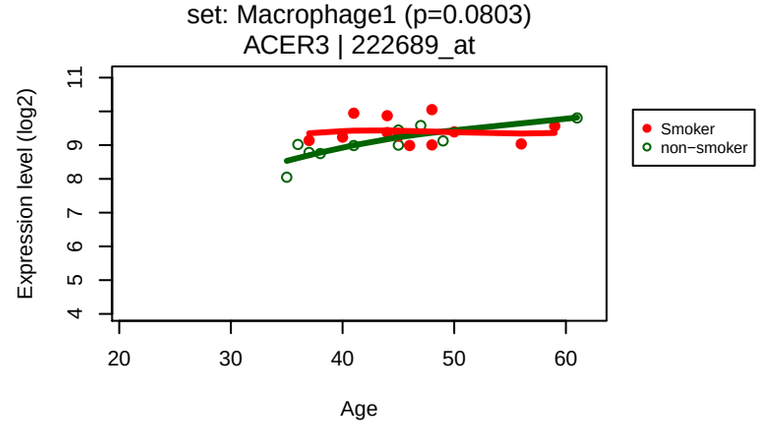

set: SmallAirway1 (p=0.3476)
ACER3 | 222689_at
Expression level (log2)
Age
Smoker
non−smoker

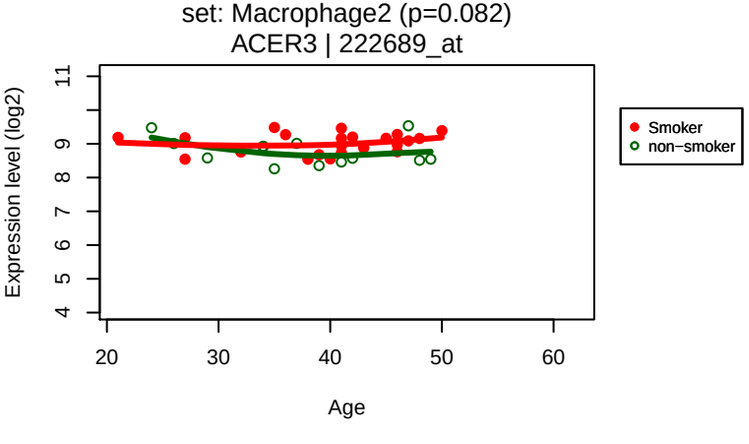

set: SmallAirway2 (p=0.0854)
ACER3 | 222689_at
Expression level (log2)
Age
Smoker
non−smoker

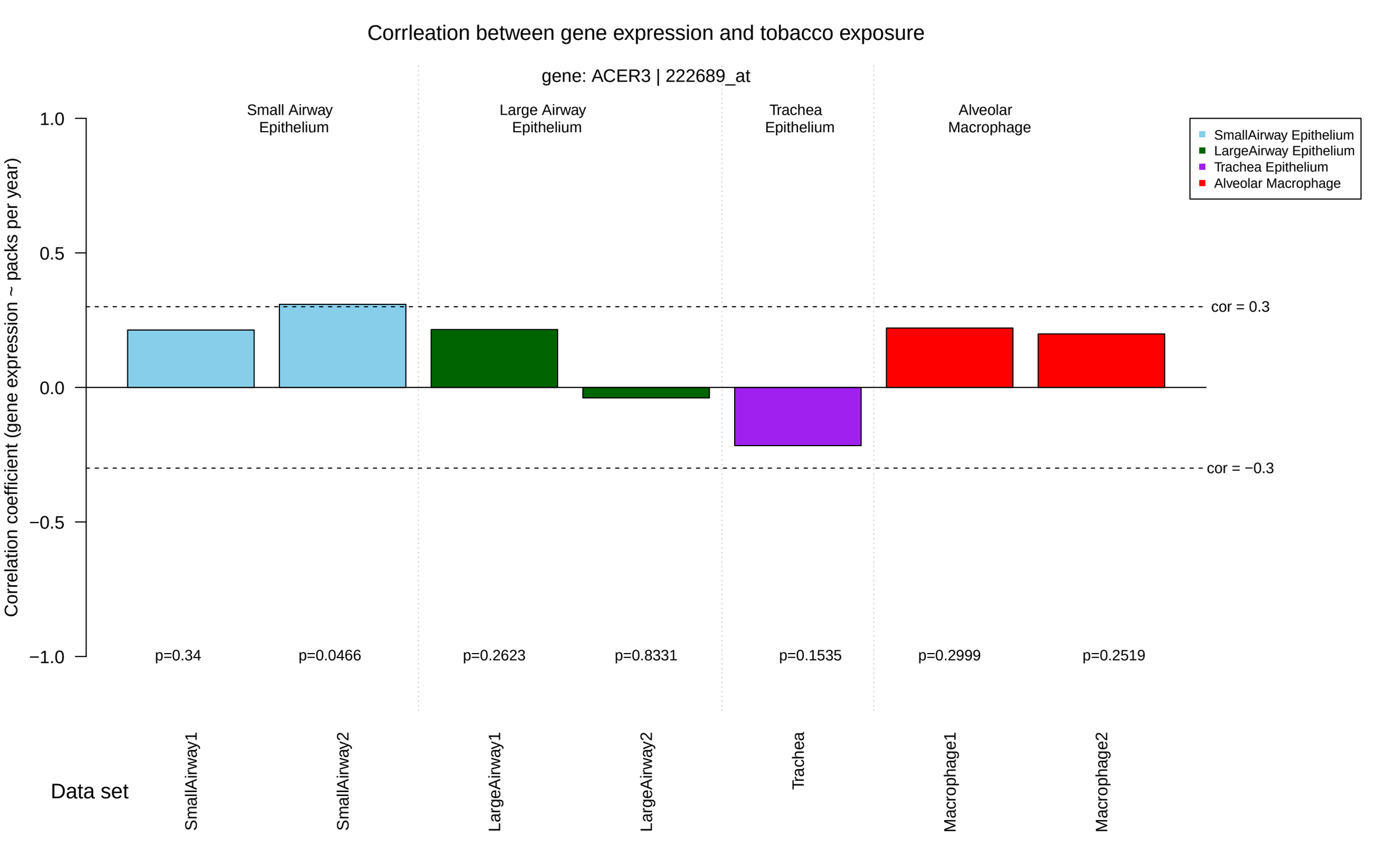

set: LargeAirway1 (p=0.6828)
ACER3 | 222689_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1542)
ACER3 | 222689_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.327)
ACER3 | 222689_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0803)
ACER3 | 222689_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.082)
ACER3 | 222689_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACER3 | 222689_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.34
p=0.0466
p=0.2623
p=0.8331
p=0.1535
p=0.2999
p=0.2519
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2